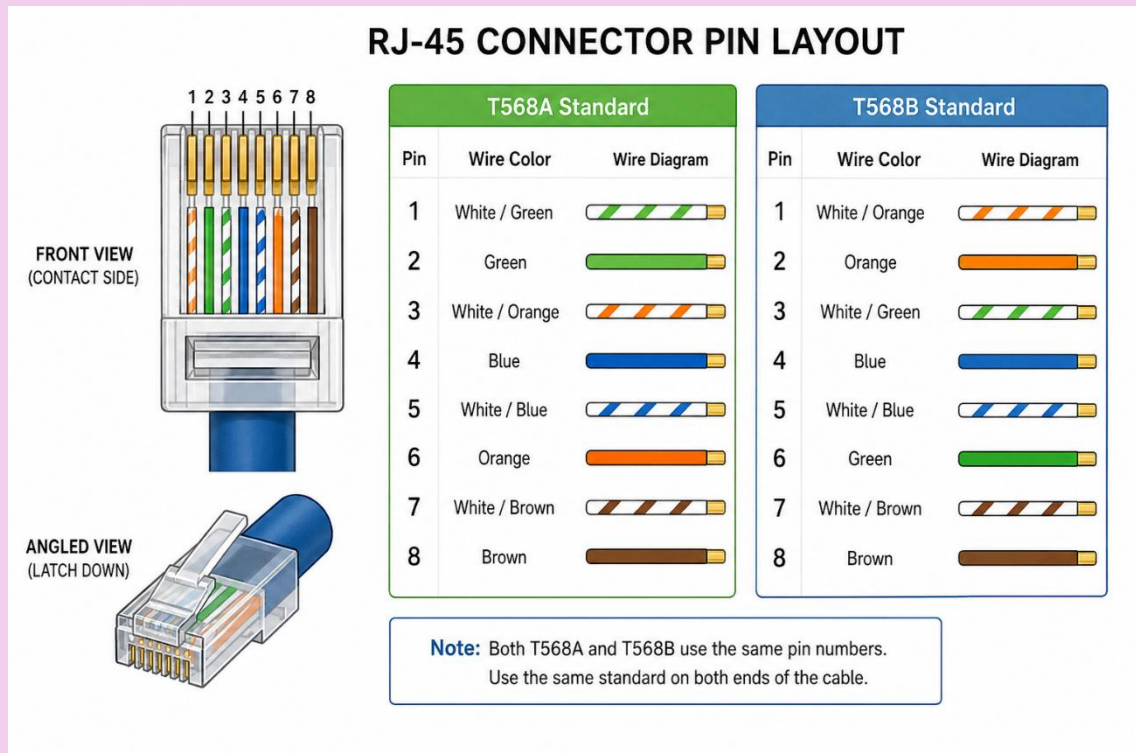


Ethernet standard and specification

1. Draw a simple diagram showing the pin layout of an RJ-45 connector and label the color codes for the T568A and T568B Ethernet wiring standards.



2. Create a table comparing the key specifications of Ethernet standards 10BASE-T, 100BASE-TX, and 1000BASE-T, including speed, cable type, maximum cable length, and connector type.

Ethernet Standard	Maximum Cable		Connector Type
	Speed	Cable Type	
	Length		
	Cat3 or higher UTP (Cat5/Cat5e/Cat6 also supported)		
10BASE-T	10 Mbps	100 m (328 ft)	RJ-45
100BASE-TX	Cat5 or higher UTP (Cat5e/Cat6		RJ-45

1000BASE-T	1 Gbps (1000 Cat5e or Cat6 UTP (Cat6 recommended 100 m (328 ft) Mbps) for better performance)	RJ-45
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100 Mbps 100 m (328 ft)
recommended)

- Given a scenario where you need to connect two laptops directly using Ethernet, explain which cable (straight-through or crossover) you would use and why.

Hint: Think about how transmit and receive pairs are wired in each cable type.
 ➤ To connect **two laptops directly using Ethernet**, you would traditionally use an **Ethernet crossover cable**.

Why a crossover cable?

Both laptops are the **same type of device (host-to-host)**. Their Ethernet ports transmit (TX) and receive (RX) on the same wire pairs. A crossover cable swaps these pairs so that:

- **Laptop 1 TX**
→ **Laptop 2 RX** •
 Laptop 1 RX
← **Laptop 2 TX**

This allows the two devices to communicate directly without needing a switch or hub.

- List three real-world devices in your home or college that use Ethernet connections, and for each, specify the likely Ethernet standard (e.g., 100BASE-TX, 1000BASE-T) they support.

Device	Typical Use	Likely Ethernet Standard	Speed
Desktop Computer	Internet access, online gaming, file sharing	1000BASE-T (Gigabit Ethernet)	1 Gbps
Wi-Fi Router	Connects the local network to the Internet	1000BASE-T (most modern routers)	1 Gbps
Network Printer	Shared printing over the network	100BASE-TX (many printers) or 1000BASE-T (newer models)	100 Mbps or 1 Gbps

- **Smart TV** – Usually **100BASE-TX** (100 Mbps), though some premium models support **1000BASE-T**.
- **Network Attached Storage (NAS)** – Typically **1000BASE-T**, with high-end models supporting **2.5GBASE-T** or **10GBASE-T**.
- **IP Camera** – Commonly **100BASE-TX**.